

# NGUYEN QUY DANG

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AI Research & Machine Learning Systems | Deep Learning | Optimization | Large-Scale Computing

## Education

### Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Aug 2026–May 2028

*B.Sc. in Computer Science; Advisor: Dr. Abdulrahman Mahmoud*

*Abu Dhabi, UAE*

- Research direction: high-performance computing, systems for machine learning, machine learning for systems, and reasoning-aware compression for large reasoning models.

### University of Engineering and Technology, Vietnam National University

Aug 2022–Dec 2025

*B.Sc. in Artificial Intelligence, GPA: 3.69/4.00 (top 10%)*

*Hanoi, Vietnam*

- Selected for the Advanced Program in Artificial Intelligence as one of 28 students from a cohort of 174; ranked 1st of 171 students in Semester I, 2024–2025.

## Research Publications

- **Nguyen Quy Dang et al.** “RE-LL1: An Effective Regularized  $(L, L, 1)$ -Tensor Decomposition Method for Video Background–Foreground Modeling and Separation.” *ICASSP 2026*. [Code](#)
- **Nguyen Quy Dang et al.** “Fast and Robust Triple Tensor Decomposition With Data Corruption.” *ICASSP 2026*. [Code](#)

## Honors & Awards

- Third Prize, ICPC Vietnam Regional (2024); Second Prize, ICPC Vietnam National (2023, 2024).
- Second Prize, Vietnam National Olympiad in Informatics (2020, 2021); Third Prize, Vietnam Student Informatics Olympiad – Super Cup Division (2025).
- Vietcombank Scholarship (2025); Odon Vallet Scholarship (2021); university merit scholarships (2023–2025).

## Research & Technical Experience

### BioTuring

Dec 2025–Aug 2026

*Software Engineer; Algorithms*

*Ho Chi Minh City, Vietnam*

- Designed a production lossless-compression pipeline for a sparse gene dataset containing **20 million cells**  $\times$  **45,000 genes**, distributed across **four NVIDIA H100 GPUs**.
- Developed **Rice coding with beta pruning** from first principles, reducing data to **10% of its original size** and achieving approximately **2 $\times$  stronger compression** than NVIDIA’s **nvCOMP ANS API** at comparable runtime.
- Implemented four-GPU **t-Digest**, **Greenwald–Khanna**, and **KLL** streaming-quantile algorithms in production, applying statistical approximation methods to large, sparse biological data.

### MOREH

Jun 2025–Oct 2025

*GPU Trainee*

*Hanoi, Vietnam*

- Developed and benchmarked performance-critical GPU training and inference paths for generative image and transformer models on NVIDIA and AMD accelerators.
- Applied CUDA/ROCm profiling, kernel optimization, and data, expert, and pipeline parallelism to improve multi-GPU utilization.

### Teko

Feb 2025–Apr 2025

*Data Scientist Intern*

*Hanoi, Vietnam*

- Fine-tuned **DeepSeek** and **LLaMA** variants and evaluated generation quality using BERTScore and METEOR.
- Structured domain data and built a Pinecone vector database for retrieval-augmented generation workflows.

### Wisdom Brain Solution

Jan 2024–Jul 2024

*AI Research Intern*

*Remote*

- Applied **Direct Preference Optimization** to transformer-based English–Vietnamese translation using EnvitT5-Translation and mBART50.
- Designed preference-pair experiments that improved BLEU by **23 points** and back-translation BLEU by **16 points**.

## Selected Research Projects

### StyleGAN Performance Optimization | CUDA, PyTorch, RTX 3090

GPU Systems

- Increased end-to-end training throughput from **0.07 to 330 images/s** (approximately **4,700 $\times$** ) on identical RTX 3090 hardware.
- Optimized Conv2D and ConvTranspose2D using operator fusion, shared-memory tiling, memory coalescing, and mixed-precision computation.
- Removed CPU input bottlenecks through asynchronous loading, optimized augmentation, and compute/I/O overlap; evaluated stability with FID sweeps.

### GPT-OSS Inference Optimization | HIP, ROCm, AMD MI250

Transformer Systems

- Increased decoding throughput from **2 to 14,500 tokens/s** (approximately **7,250 $\times$** ) on identical AMD MI250 hardware.
- Optimized attention using flash-style kernels, blockwise softmax, and contiguous KV-cache layouts; fused bias, activation, and epilogue operations in the MLP.
- Tuned streams, occupancy, and memory access while applying data, expert, and pipeline parallelism for multi-GPU execution.

### Federated Learning Under Data Heterogeneity | PyTorch, ViT, Statistical Testing

GitHub

- Investigated FedAvg, FedProx, and FedNova under balanced, weakly heterogeneous, and strongly skewed non-IID MRI client distributions on datasets up to **60 GB**.
- Compared CNN, ResNet, VGG, InceptionNet, and **Vision Transformer** models using accuracy, F1, AUC, paired *t*-tests, DeLong's test, McNemar's test, and permutation tests.
- Built an open-source simulation and visualization platform; observed stronger ViT resilience under high variance and improved stability with FedNova under imbalanced updates.

**Visual Language DENSO Expert** | GPT-4o, Retrieval, Whisper, Diffusion

**Second Runner-up**

- Built a multimodal factory-support system combining GPT-4o vision-language reasoning with retrieval over domain documentation and Whisper-based speech interaction.
- Used diffusion models to generate synthetic training data and placed Second Runner-up at the 2024 DENSO Hackathon.

**Multi-Agent Reinforcement Learning** | Self-Play, Policy Optimization

**GitHub**

- Implemented self-play for MAgent2 battles and designed convolutional policy and value networks to model local spatial interactions among agents.
- Trained agents with a clipped policy objective that constrained large updates while balancing exploration and exploitation.

## Technical Skills

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**Programming & Systems:** Python, C++, CUDA C/C++, HIP, SQL; CUDA, ROCm, NCCL/RCCL, distributed and multi-GPU computing

**Machine Learning:** PyTorch, Hugging Face, transformers, LLM fine-tuning, DPO, reinforcement learning, federated learning

**Research Methods:** optimization, probability and statistics, tensor decomposition, streaming algorithms, statistical testing

**Tools:** Linux, Git, Docker, NumPy, Pandas, scikit-learn, Nsight Systems/Compute, PyTorch Profiler, rocprof